How We Pay Matters?

Payment Modality and Spending Behavior in Cash Transfer Programs

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Project Question: How does receiving cash transfers through mobile wallets, compared to physical cash, affect household spending patterns in Indonesia?

Introduction

This research project addresses the intersection of digital financial inclusion and social protection policy, specifically examining how the delivery of cash transfer programs influences household spending behavior. As governments worldwide increasingly digitalize social assistance programs, understanding whether the method of transfer delivery (mobile wallets or physical cash) affects how recipients allocate resources has significant implications for program design and poverty alleviation outcomes.

The shift to digital transfers matters for several reasons. First, mobile money transfers may promote more spending. Discussions about mobile money in developed countries have focused on the "cashless effect," an increased willingness-to-pay (WTP) induced by using cashless forms of payment. The theory suggests physical forms of money are psychologically more salient due to their observability while digital forms are less salient. Using e-money lowers the "pain of paying," especially when user friction is minimized in cases of tapping a credit card or scanning a QR code. In this theory, salience, "pain of paying," and convenience are mediators explaining why e-money would cause a higher WTP.[1] However, in environments where technological literacy is low, it is unclear whether e-money would facilitate higher spending given low adoption and low comfort using mobile phones.

Second, mobile money transfers in lieu of cash could reduce transaction costs and waiting times associated with receiving the fund. An unconditional cash transfer program in Niger found that when transferring mobile wallets compared to enveloped-cash, recipients forwent the transport costs required to collect the cash, leading to better time allocation and increased consumption of protein-rich foods.[2]

Third, digital payments could alter spending patterns by creating psychological framing effects. Specifically, money held in a mobile wallet may be perceived or spent differently than physical cash.[3] The built in recordkeeping features of many mobile wallets (including transaction history and current balance) may encourage frugal spending compared to cash. In a similar vein, money that must remain liquid but is set aside for short to medium term goals may be deposited as savings in a mobile account, as found in the Kenyan financial diaries.[4] Mobile wallets also easily facilitate gift giving and remittances, especially when the recipient is far away. These distinct use cases for mobile money may not be possible with cash, and so digital cash transfers may promote these behaviors.

Indonesia provides an ideal setting for research to better understand this causal chain. The country operates one of the largest social assistance programs in the developing world, with cash transfers reaching over 10 million households through programs like Program Keluarga Barapan and Bantuan

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[1] Boden, J., Maier, E., & Wilken, R. (2020). The effect of credit card versus mobile payment on convenience and consumers' willingness to pay. Journal of Retailing and Consumer Services, 52, 101910. https://doi.org/10.1016/j.jretconser.2019.101910

[2] Aker, J. C., Boumnijel, R., McClelland, A., & Tierney, N. 2016. Payment mechanisms and antipoverty programs: Evidence from a mobile money cash transfer experiment in Niger. Economic Development and Cultural Change, 65(1), 1–37. https://doi.org/10.1086/687578

[3] Thaler, Richard H. 1999. "Mental Accounting Matters." *Journal of Behavioral Decision Making* 12 (3): 183–206. https://doi.org/10.1002/(SICI)1099-0771(199909)12:3%3C183::AID-BDM318%3E3.0.CO;2-F.

[4] Citation needed

Pangan Non Tunai.[5] Simultaneously, the Indonesian government has been progressively shifting toward digital payment mechanisms as part of its broader financial inclusion agenda.[6]

Currently, Indonesia's cash transfer programs use mixed delivery mechanisms: some beneficiaries receive physical cash at distribution points, others receive transfers through bank accounts, and an increasing number receive funds via mobile wallets.[7] While existing research has examined cash versus in-kind transfers and mobile money adoption separately, there is limited experimental evidence on how cash transfer delivery channels affect spending behavior, particularly in Southeast Asia. This evidence gap is critical: if digital delivery mechanisms systematically alter spending patterns, this would have important welfare implications that should inform social protection program design.

Our research question is as follows: How does receiving cash transfers through mobile wallets, compared to physical cash, affect household spending and allocation in Indonesia.

The key variables to assess the research question as follows:

- **Treatment Variable (X):** Delivery mechanism of cash transfers (mobile wallet vs physical cash).
- **Outcome (Y):** Differences in household expenditure patterns measured as:
    - Total expenditure
    - Total expenditure levels by category (food, energy, communications, transportation, cleaning materials, water, education, personal care, religious, alcohol/cigarettes/drugs, medical, clothing, housing, other)
- **Population:** Low-income households in Indonesia

Background

Indonesia is a large and geographically diverse country. As the world's fourth most populous country, Indonesia had an estimated 284.4 million people in 2025, and it remains an archipelagic state of more than 17,000 islands, creating persistent challenges for service delivery, financial access, and benefit distribution across remote and rural areas. 23.36 million people, or 8.25 percent of the population, are living below the national poverty line as of September 2025.[8] Figure 1 geographically presents the province level average poverty rates for Indonesia in September 2025.

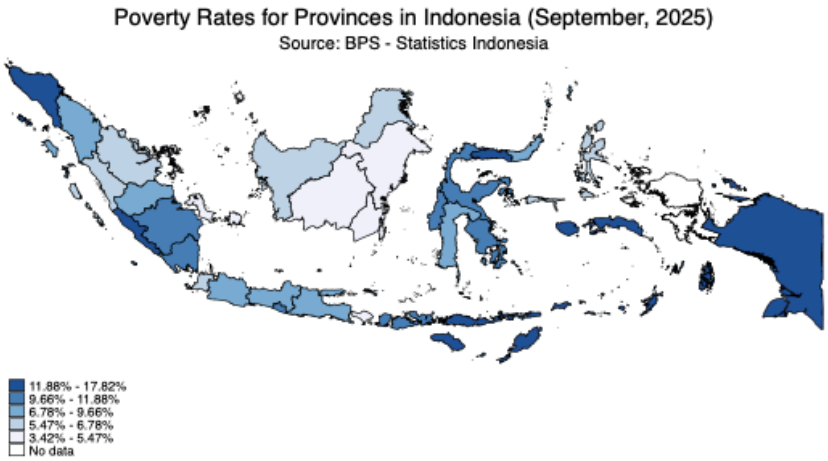
Figure 1: Poverty Rates for Provinces in Indonesia

[5] World Bank. 2019. Aspiring Indonesia—Expanding the Middle Class. World Bank, Washington, DC. https://doi.org/10.1596/33237.

[6] Darjana, and Yanti Rusmawati. 2025. "Performance of Digital Economic Indicators During the COVID-19 Pandemic: Payment System Case of Indonesia." Bulletin of Monetary Economics and Banking 28, no. 4: 595–608. https://bulletin.bmeb-bi.org/cgi/viewcontent.cgi?article=2513&context=bmeb.

[7] Palaon, Hilman. 2020. "Digitizing G2P Payments: Lessons from Indonesia." CGAP (blog). July 21, 2020. https://www.cgap.org/blog/digitizing-g2p-payments-lessons-indonesia.

[8] The poverty rates are obtained from BPS-Statistics Indonesia. The data contains poverty rates for urban, rural, and both area across provinces in Indonesia. Urban area has the mean rate of 6.55%, while rural area has 13.35% with the max value of 39.14%. Papua province has the highest total poverty rates of 29.45%.



Indonesia's transition away from cash has unfolded gradually through successive policy efforts, including the launch of the Gerakan Nasional Non Tunai (GNNT) and the reform of the G2P program, "G2P 4.0." GNNT launched in 2014 is a broad national push toward non-cash payments by Bank Indonesia to encourage a shift from cash-based transactions toward non-cash payments systems as a broader inclusive digital economy agenda. Although the digitalization of payment systems under GNNT has accelerated the adoption of non-cash transactions,[9] important barriers remain. Infrastructure gaps and low digital literacy in non-urban areas continue to be a hurdle for financial inclusivity.

"G2P 4.0" refers to Indonesia's G2P digitization reform to move beyond simple payment digitalization toward a broader beneficiary-centered architecture that emphasizes efficiency, targeting accuracy, governance, interoperability, and financial inclusion. The reform has been successful in establishing a national roadmap of G2P digitization, but also left major challenges, including incomplete implementation of beneficiary choice, uneven rural access and agent networks and limited financial capability among recipients (Palaon, 2020).

Regarding the mobile money ecosystem, Indonesians can choose from a range of apps as their primary e-wallet. These apps differentiate themselves by providing distinct services, including ride sharing, food delivery, online shopping, discounted goods and banking services, and loyalty points. Indeed, e-wallet apps are more than just e-wallets; they have become "Super apps," which have many features in one app. Popular providers include GoPay, OVO, DANA, and LinkAja.[10]

The numerous services these apps provide may be overwhelming to users, especially to those with low levels of technological literacy. This is reflected in the widespread prevalence *warungs*, small businesses and kiosks that facilitate e-money transactions. Many physically hand their phones to a *warung* to complete a purchase or withdrawal. According to the 2025 FinDev report, only about 30% of adults in

[9]Faiqoh, D. N., Riyadi, S., Iswati, H., Jemmy, J., & Sugiarto, D. (2025). Implementation of the National Non-Cash Movement (GNNT) program in the digital era with assistance from local financial institutions and agencies to improve the performance of MSMEs in non-urban regional markets. Economics and Social Analytics Journal (ESA), 4(4). https://doi.org/10.55927/esa.v4i4.92

[10] Fintech News Indonesia. 2026. "5 Top E-Wallets in Indonesia and What They're Best For (2026)." February 12, 2026. https://fintechnews.sg/5-top-e-wallets-in-indonesia/.

Indonesia can independently perform a transaction without assistance despite 80% owning a mobile phone.[11]

Even still, the Indonesian government has leveraged mobile money platforms in their social programs and tax collection. Kartu Prakerja is a job training program which pays cash in exchange for individuals to sign-up for online courses. The transfer could be distributed through one of four mobile wallets.[12] As of 2026, the Ministry of Finance now collects transaction data from e-money platforms for tax collection.[13] Given the Indonesian government's integration of financial technology and commitment to digitize, Indonesia would benefit from research on the impacts of cash transfer modality.

Intervention

To estimate how cash transfer modality affects spending patterns, we will administer our own unconditional cash transfer (UCT). Eligibility is determined using criteria adapted from the BLT Kesra program, targeting households in the bottom four income deciles as identified by a proxy means test. This test incorporates indicators of asset ownership, housing characteristics, household dependency ratios, and geographic location.

Households must not currently receive PKH or Sembako transfers to be eligible for the project. Excluding current program beneficiaries avoids contamination with existing transfer income, allowing us to cleanly isolate the effect of payment modality. Households must have at least one adult member with access to a mobile phone with e-money app installed.

Participants will be randomly assigned to one of three experimental arms: (1) a control group that receives no transfers, (2) a treatment group receiving transfers in physical cash, and (3) a treatment group receiving transfers via mobile money. Both treatment arms receive transfers of identical value and frequency, differing only in the modality of delivery.

Treatment groups will receive IDR 150,000 (~$9 USD) on a biweekly basis for two years. In addition, enumerators will conduct biweekly household visits during non-transfer weeks to collect data on spending patterns. The two year project timeline will control for seasonal differences in income and spending patterns, and the biweekly survey schedule will minimize recall error among participants.

We will stratify our sampling to control for urban and rural differences. Based on BPS' classifications, rural and urban regions will be defined based on population density, percent of households with primary occupation being agriculture, and access to urban facilities (hospitals, paved roads, schools, and markets).[14] Stratifying in this way ensures that our experiment is representative of Indonesia's population and accurately captures the outcome variable.

After households have been selected, we will administer a baseline survey which collects demographic information on the household (housing conditions and ownership, household size, ages of household members, education attainment), income sources, and assets (property, vehicles, livestock, etc.). This

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[11]Shinozaki, Shigehiro. 2024. "As Digitalization Grows, Will Indonesia's Agent Networks Survive?" *FinDev Gateway* (blog). May 22, 2024. https://www.findevgateway.org/blog/2024/05/digitalization-grows-will-indonesias-agent-networks-survive.

[12] UN DESA. 2021. "Program Kartu Prakerja (Pre-Employment Card Program)." Department of Economic and Social Affairs. https://sdgs.un.org/partnerships/program-kartu-prakerja-pre-employment-card-program.

[13]Tech in Asia. 2024. "Indonesia to Access E-wallet, Crypto Data Under 2025 Tax Rule." November 15, 2024. https://www.techinasia.com/news/indonesia-to-access-e-wallet-and-crypto-data-under-2025-tax-rule

[14]Arisandi, D., and A. Syakir. 2023. "Analysis of Using Google Maps Data to Measure the Presence or Accessibility of Urban Facilities for BPS - Statistics Indonesia Classification of Urban and Rural Villages." *Journal of Information Systems and Informatics* 5 (1): 308–321. https://doi.org/10.51519/journalisi.v5i1.442.

baseline is essential for verifying balance between treatment and control groups, establishing pre-intervention spending shares across categories, and supporting difference-in-differences estimation. Following the two-year diary period, we plan to conduct a follow-up survey approximately six months after the final diary visit, designed to capture any persistence or fade-out of treatment effects beyond the active intervention window.

Primary Data: Household Financial Diaries

Our primary data collection strategy is modelled on the financial diaries approach, a high-frequency survey methodology that tracks every income and expenditure transaction a household makes over an extended period. Unlike one-shot recall surveys, financial diaries capture the timing, volume, and composition of spending in granular detail, making them well-suited to detecting shifts in spending behaviour across categories, which is precisely what our outcome requires.

At each visit, respondents will be asked to recall all transactions since the previous visit across the following expenditure categories:

- Food
- Energy
- Communications
- Transportation
- Cleaning materials
- Water
- Education
- Personal care
- Religious
- Alcohol/cigarettes/drugs
- Medical
- Clothing
- Housing
- Other

For each transaction, we will record the amount spent, the payment method used (cash, mobile wallet, or other), and where the purchase was made. Additionally, we will collect data on income, assets (including outstanding debts and savings), and "going-ons" (major life updates or challenges which may impact financial health). We will also collect data on intra-household control over the transfer and whether the modality affects which household member manages and spends the money. This is particularly relevant for female-headed or male-dominated households given evidence from similar programmes that digital transfers can shift bargaining power toward women.[15]

In-person administration is preferred over self-completion for several reasons: it reduces recall error, allows enumerators to probe and clarify ambiguous transactions, builds trust with respondents over time, which is important for sensitive categories like alcohol or religious giving, and ensures consistent data quality across treatment and control arms. Additionally, consistent in-person visits allow for the collection of valuable qualitative data which provides insights on how mental accounting differs between the control and treatment arms.

[15] Aker, J. C., Boumnijel, R., McClelland, A., & Tierney, N. 2016. Payment mechanisms and antipoverty programs: Evidence from a mobile money cash transfer experiment in Niger. Economic Development and Cultural Change, 65(1), 1–37. https://doi.org/10.1086/687578

Secondary Data: BPS Administrative Records
To complement primary data, we will draw on administrative data from BPS, specifically the household expenditure and welfare statistics tables available through the BPS social protection database. This data provides regional and national benchmarks for household spending patterns among low-income populations, allowing us to assess whether our recruited sample is broadly representative of the target population and to contextualise observed treatment effects against wider trends.

Data Quality Challenges
Several data quality risks warrant attention. The most significant is attrition. Over a two-year period with fortnightly visits, households may move, withdraw consent, or become unreachable, particularly in rural areas or among more mobile members of the sample. Attrition is especially concerning if it is differential across treatment and control arms, for instance, if mobile wallet recipients who struggle with the technology disengage from the study at higher rates, our estimates would be biased. To mitigate this, we will track reasons for attrition carefully and test whether attrition is random or systematic using baseline characteristics.

A second concern is survey fatigue. Fortnightly visits over two years are demanding for respondents. We will monitor response quality over time and build in small non-cash incentives, such as airtime top-ups, to maintain engagement, taking care that these do not themselves alter spending behaviour in ways that contaminate our outcomes.

Third, recall bias is inherent in any diary methodology that relies on retrospective reporting within a two-week window. Enumerators will be trained to use structured prompts and reference to regular versus irregular spending to aid recall, and we will flag categories with likely underreporting, such as alcohol and tobacco, for cautious interpretation.

Finally, selection bias in enrolment is a concern if the household screening process does not reach the full population of eligible households. Households that are harder to contact, more geographically remote, or less trusting of government-adjacent surveys may be systematically underrepresented. We will document recruitment rates by district and compare the characteristics of enrolled versus non-enrolled eligible households where possible.

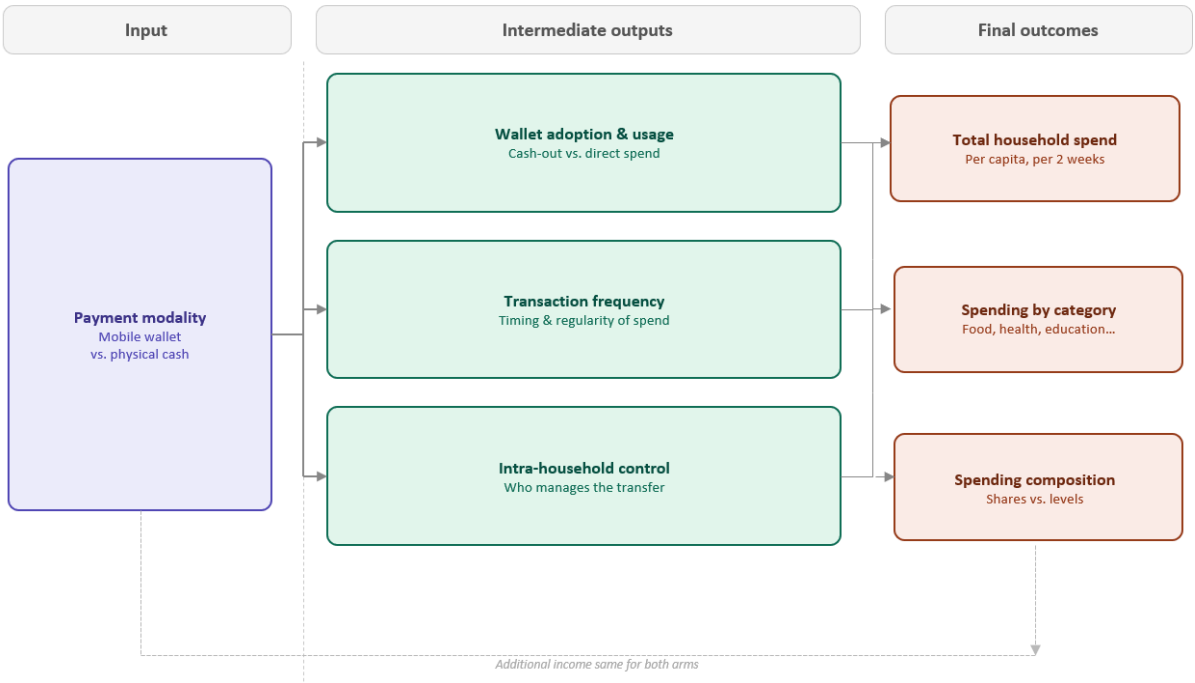
Hypotheses and Outcome Construction
Our primary outcome is total household expenditure per capita in the two weeks following each diary visit, and the share of that expenditure allocated to each of the categories listed above. We will construct these as both levels (in Indonesian Rupiah, IDR amounts) and shares (proportion of total spending) to allow us to distinguish between effects on overall spending volume and compositional shifts across categories.

Our primary hypothesis is that households receiving transfers via mobile wallet will show different spending patterns compared to those receiving physical cash, but the direction is theoretically ambiguous given our two competing mechanisms. Under the low-friction / cashless effect pathway, more likely in urban settings with high QRIS merchant acceptance (QRIS is a unified QR code payment system introduced by the Bank of Indonesia that works across all e-wallets), we would expect mobile wallet recipients to spend more overall, with increases concentrated in categories that are accessible digitally such as communications, food, and clothing.

Under the high-friction pathway (likely present in rural areas where technological literacy is low and access to QRIS vendors is diffuse), we expect changes, though in which categories is uncertain. For those who prefer cash over e-money, we may expect spending on transportation and withdrawal fees to rise as individuals travel to convert their mobile funds to physical cash. Alternatively, the “pain of paying” in the

high-friction path might deter discretionary spending. We may also record higher rates of remittances or gift giving, as e-money can connect users to faraway recipients; giving money through mobile wallets may nudge people to contribute to their social networks.

Secondary outcomes include looking at spending in specific categories where we have clearer expectations about what might change. Food spending is likely to go up regardless of which pathway plays out, which is consistent with findings from similar programmes in Niger and Uganda. Spending on medical care and education may increase if mobile wallets give recipients, particularly women, more control over how money is used within the household. Savings may increase if participants find mobile money to be a viable savings instrument. Finally, spending on communications is likely to rise in the mobile wallet group simply because having a wallet makes it easier and cheaper to top up mobile credit and pay phone bills directly.



Power Calculations

As this experiment has three arms, no transfer, physical cash, and mobile money, the power is calculated for pairwise comparisons, as Gupta and Kopper (2021) suggests.

- Cash vs mobile money
- Cash vs control
- Mobile money vs control

We ensure that the study is powered for the smallest MDE among the comparisons that will be made, and apply adjustments for multiple hypothesis testing, such as with the Bonferroni or Benjamini Hochberg adjustments, consistently across both power calculations and analysis.[16]

[16]Gupta, S., Kopper, S. (nd). "Power Calculations." J-PAL. https://www.povertyactionlab.org/resource/power-calculations.

Components of power calculations include:

- Significance (α): 5%.
- Power (1- β): 80%
- MDE: 5%
- Attrition: 10%
- Sample size (N)
- Variance of outcome variable (σ^2):
- Treatment allocation (P): 33%
- Intra-cluster correlation coefficient (ICC): 0.4

We can reasonably expect 10% of the sample to attrite. The sample period is long and the commitment level is high. We hope that our non-cash incentives, such as airtime top-up, will keep participants in the survey.

We predict that ICC will be 0.4. By stratifying our sample between urban and rural, we in effect create two clusters. The ICC is the measure of within-cluster correlation. We expect ICC to be low since although being a rural or urban area may predict economic outcomes, it does not have a high explanatory power.

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